

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1856

Donner 138 kV
March 26, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1856. The Interconnection Customer proposes a 171.5 MW natural gas combined cycle facility interconnecting at the Donner 138 kV substation in the Delaware area. The request was received on August 13, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 531 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,819,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
LONGMIRE - ORANGE BULK 138kV	DONNER - NINEMILE PT 138kV	158.2	196	64.3
DONNER SUBSTATION - ORANGE BULK 138kV	DONNER SUBSTATION - NINEMILE PT 138kV Line	111.4	291	100.0
DONNER - NINEMILE PT 138kV	NINEMILE PT - LONGMIRE 138kV	103.7	477	77.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LONGMIRE - ORANGE BULK 138kV	Rebuild line with higher-capacity conductor	\$6,357,000	41% (\$2,627,000)
DONNER - ORANGE BULK 138kV	Rebuild line with higher-capacity conductor	\$26,257,000	45% (\$11,794,000)
DONNER - NINEMILE PT 138kV L	Reconductor line	\$979,000	41% (\$398,000)
	Total allocated to project		\$14,819,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.